

**EFFECT OF SUNLIGHT EXPOSURE IN MICROBIAL QUALITY STANDARDS IN
BOTTLED DRINKING WATER UNDER AMBIENT ENVIRONMENTAL
CONDITIONS COLOMBO DIVISIONAL SECRETARAIT**

Fathima Rukaiya Safeer

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Abstract

Consistently, there are endless morbidity and mortality because of the utilization of unsafe drinking water. Use of bottled water in Sri Lanka has increased over the last decade therefore surveillance of bottled water manufacturing industries needs to be done regularly by regulatory agencies. The present research will be carried out to assess algae analysis of bottled water exposed to sun through isolation by Sabouraud agar culture and to assess the types of bacterial species which contaminate bottled water exposed to sunlight through isolation by MacConkey agar culture in the local market of Colombo divisional secretariat. List of bottled water brands within the Colombo divisional area would be prepared and 5 brands will be selected using a random number generator and 2 bottled from each brand, once in two weeks for two months will be purchased from the manufacturer. The first batch of samples will be analysed immediately after collection and this will be considered the control batch. The second batch will be tested after 14 days, one set will be exposed to sunlight for 9 hours each day, while another set of samples will not be exposed to sunlight. The samples will be tested for the presence of bacteria using MacConkey agar culture method and will be incubated at 35 °C for 24 h. Algae analysis will be done by Sabouraud dextrose agar culture methods and will be incubated at 25-30 °C and examined twice a week for 4-6 weeks. The colonies of agar plates will be counted and expressed as colony forming units per mL (cfuml⁻¹). For bacterial identification colony morphology and gram stain method will be used while algae identification will be done using the compound microscope. Results obtained will be analysed using chi squared and T-test method and compared with WHO water quality standards of bottled water.

Effect of sunlight exposure in microbial quality standards in bottled drinking water under ambient environmental conditions in Colombo Divisional Secretariat

1.0 Introduction

1.1 Background Information

Water is a fundamental prerequisite of all living things. Acceptable source of hygienic, safe, and sterile drinking water is vital for wellbeing. Access to safe drinking water is a crucial operator in human life. Inaccessibility of good quality drinking water is broad and this leads to health consequences. Diseases caused due to unhygienic water are one of the prime health topics worldwide (Shams *et al.*, 2019). There is a high frequency of the diarrhoea among adolescents and new-born children that can be trailed to the utilization of hazardous water and unsanitary practices. Continuous increase in the sale and indiscriminate consumption of packaged drinking water is of public health significance. Hence, to protect community wellbeing, bottled water must be of the high quality (Venkatesan, Balaji and Victor 2014).

Consumption of bottled water continues to increase steadily as a result of public concerns about palatability and microbial and chemical contaminants in tap water. In 1995, 2.7 billion gallons of bottled water were consumed in the U.S. European consumption of bottled water has increased 200% in the past decade and developing nations such as China and Indonesia are projected to increase demand by 30% in the next 5 years (Grant, 1998). Severe guidelines have been enforced to regulate the quality of bottled water in several countries because it has become prominent throughout the world. The World Health Organization (WHO) and the European Economic Community (EEC) have suggested general rules for the quality of water utilized by humans (Owamah, 2019). As indicated by these instructions *E. coli* or thermotolerant coliform organisms must not be detected in a 100 mL test (Codex standard 2001) while bottled water quality specifics approved heterotrophic plate count (HPC) limits 50 colony units each mL (cfu/mL) (WHO 2002). Chemical parameters also take up responsibility in the quality of drinking water. Specifically, trace metals and few anionic elements, for example, nitrates, phosphates, etc frequently give significant insights for the source water contamination. A few past examinations have demonstrated that chemical quality constituents of bottled waters are not reliable with the prescribed levels (Al-Saleh and Al- Doush 1998; Ozturk and Yilmaz 2000).

Use of bottled water in Sri Lanka has been increasing over the last decade. Bottled water is popular among most consumers, especially those traveling and on health care. In emergencies such as natural disasters, bottled water is used extensively to supply safe water. Because of this high demand, the water bottling industry has been booming throughout the country. Every year an increase in sales of bottled water is seen not only in Sri Lanka, but also globally (Penland and Wilhelmus 1999). In the last decade usage of bottled drinking water in Sri Lanka has been expanding. It is most prominent amongst shoppers, particularly health care and journeying. During situations like catastrophic events, bottled water is utilized broadly to provide safe water. Due to this high demand, the bottled water industry has been prospering all through the nation and worldwide (Sasikaran *et al.*, 2012).

The Public Health Division of the Ministry of Health, Sri Lanka 2019, registers 181 packaged drinking water brands. Packaged and natural mineral water makers producers get a verification of registration from the Food Authority of Sri Lanka, to either produce or import bottled drinking water. Moreover, because of expanded interest, there is a theory that numerous unlawfully packed bottled water brands are present in the market. The troubling details of the story are that already utilized bottles from unsafe places are being refilled (Herath *et al.* 2012). Thusly the primary purpose of this investigation is to survey the microbiological nature of bottled water in the local market and to assess whether the quality parameters meet the allowed levels, and consequently make awareness of the consequences bottled drinking water in Sri Lanka.

1.2 Justification

According to a study conducted by Diduch, Polkowska and Namieśnik (2012) contamination in bottled water is increasing and this is a complex issue, because the quality of bottled water is the resultant of various variables, beginning with its primary conformation, through a packaging procedure, to the conditions wherein the product is stored. The continually rising usage of bottled water and consumer's belief that they are of high quality shows the need to emphasize on this topic (Joseph *et al.*, 2018). Hence, periodic surveillance bottled water is essential. This will serve the dual purpose of monitoring the standards of bottled water production industries as well as help in quality. Use of bottled water in Sri Lanka has increased over the last decade, while new brands of bottled water are often introduced to the market. However, the manufacturers' adherence to bottled water regulations is questionable, raising concerns regarding the quality of bottled water in the country. In a previous study conducted in Sri Lanka eighty percent of brands showed higher heterotrophic plate counts (HPC) which exceeded the WHO guidelines for bottled drinking water (Herath *et al.*, 2012). The outcomes of this research on bottled water quality allows an enhanced understanding of the situation which is damaging the nature of bottled water and are valuable data for shoppers. It will likewise give a strong basis by which modifications of guidelines administering bottled water quality can be based.

1.3 Objectives

1.3.1. General objective:

- To assess sunlight exposure in microbial quality standards of bottled water distributed in Colombo Divisional Secretariat under ambient environmental conditions.

1.3.2. Specific Objective:

- To assess types of algae present in bottled water exposed to sunlight in Colombo Divisional Secretariat.
- To assess the bacterial species present in bottled water exposed to sunlight in Colombo Divisional Secretariat.
- To determine the average number of viable colony counts isolated from each brand of bottled water.

2. Materials and Methods

2.1. Study design

A descriptive cross-sectional study design.

2.2. Sample Selection

A survey will be done among leading markets in Colombo divisional secretariat to select the commonly used bottled water brands. A list of leading supermarkets and convenience stores within the Colombo Divisional Secretariat will be made and 5 supermarkets and 5 convenience stores will be chosen through Simple random sampling among them to do the survey.

2.3. Study setting

Samples of water bottle will be purchased from 5 selected manufacturers located within the boundaries of Colombo divisional secretariat where bottled water is not exposed to sunlight.

2.4. Study period

Study will be conducted within the period of 9 months starting from August 2019 until May 2020.

2.5. Sampling technique

Simple Random Sampling would be used to specify the brands which will be selected for the study. Lists of commonly used bottled water brands within the Colombo divisional secretariat would be prepared and 5 brands will be selected using a random number generator.

2.6. Sample implementation

2.6.1. Collection of samples

Bottled water brands of 4 different batches of 5 different brands will be collected once in two weeks for two months directly from the manufacturing company.

2.6.2. Sample preparation

The microbial content of the first batch of samples will be analysed immediately after the collection from the manufacturer and from this initial colony counts will be taken. While the second batch will be tested after 14 days, one set will be exposed to sunlight for 9 hours per day between 8 a.m. to 5 p.m., while another set of samples will not be exposed to sunlight which will be considered as the control. Microbial assessment of the bottled water samples will be investigated with the increasing sunlight exposure at Management and Science Institute (MSI) student Biomedical Laboratory.

2.6.2.1. Bacterial Assessment

The weight of 49.53 g of dehydrated medium will be suspended into 1000 ml purified/distilled water and heated till all the medium is dissolved completely. This will be sterilized by autoclaving at 15 lbs pressure (121 °C) for 15 minutes and then cooled to 45-50 °C.

Sampled bottled water will be serially diluted (10 folds) using 9 ml of sterile distilled water against 1ml of the sample. Thereafter, 19 ml of already prepared MacConkey will be poured into the petri-dishes and gently swirled so that the medium will be evenly distributed. The

MacConkey agar will be incubated at 35 °C for 24 h. The colonies of agar plates will be counted and expressed as colony forming units per mL (cfuml⁻¹) at the MSI student Biomedical Laboratory.

2.6.2.2. Algal Assessment

The weight of 65 g of dehydrated medium will be suspended into 1000 ml purified/ distilled water and heated till all the medium is dissolved completely. This will be sterilized by autoclaving at 15 lbs pressure (121°C) for 15 minutes and then cooled to 45-50 °C.

Sampled bottled water will be serially diluted (10 folds) using 9 ml of sterile distilled water against 1 ml of the sample. Thereafter, Sabouraud dextrose agar (SDA) will be poured into the petri-dishes and gently swirled so that the medium will be evenly distributed. The SDA will be incubated at 25-30 °C and examined twice a week, Monday and Thursday, for 4-6 weeks. The colonies of SDA will be counted and expressed as colony forming units per mL (cfuml⁻¹) at the MSI student Biomedical Laboratory

2.6.3. Isolation & Identification

2.6.3.1. Test for Bacterial Identification

2.7.1.1. Colony morphology

‘Cowan and Steel’s Manual (Cowan, 1974) will be referred for colony characterization purposes.

2.7.1.2. Gram stain

To determine, whether the organisms isolated belongs to the category of gram negatives (pink colour) or gram positive (crystal violet colour) organisms.

With a sterile cooled loop, a drop of sterile water or saline solution will be placed on the slide. The loop will be sterilized again and a very small sample of bacterial colony will be picked up and gently stirred into the drop of water/saline on the slide to create an emulsion. Allow the smear to air dry and thereafter, the slide will be held at one end and passed through the flame of a Bunsen burner two times with the smear side up. To stain the slide, the smear will be flooded with crystal violet and left for 1 minute. Thereafter it will be tilted and gently rinsed with tap water. The same procedure will be followed for gram’s iodine. To decolorize, the slide will be tilted slightly and ethyl alcohol or acetone will be applied drop by drop for 5-10 seconds, until it runs almost clear and immediately rinsed with water. Finally, will be flooded with safranin to counter-stain and let stand for 45 seconds and then will be gently tilted and rinsed with tap water. Will be blot dried and will be viewed under microscope.

2.6.3.2 Test for Algal Identification

The colonial features of the fungal colonies will be elucidated along with the morphological features of the fungi using the compound microscope. ‘Illustrated Genera for Imperfect Fungi’ (Barnett and Hunter, 1998), will be referred for the identification of fungi.

2.7. Data analysis

Data will be entered into IBM SPSS Statistical Software Package 25. Then descriptive statistics will be analysed using percentages, frequencies, mean, medians and standard deviation. Chi square and t-test will be used to compare means of the number of viable colony counts in water samples with different duration of sun light exposure.

3. References

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ANNEXURE I

Time frame

Gantt chart

[illegible]

ANNEXTURE II

Budget

Items Estimated	Cost (LKR)
Transportation	3500.00
Stationery	2500.00
Printing	1500.00
McConkey Agar	7850.00
SDA Agar	8800.00
Water Bottles	6000.00
Distilled water	1000.00
Miscellaneous	1000.00
Total	27150.00